The Ising perceptron at nonzero margin: certified bounds along the capacity curve

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Abstract

The storage capacity of the Ising perceptron at margin κ is predicted by a replica-symmetric variational problem. Huang proved that the predicted threshold $\alpha_\star(\kappa)$ is an upper bound at every margin, provided four numerical conditions on explicit low-dimensional problems hold at that margin. We verify the four conditions in ball arithmetic at the margins $\kappa = -0.45$, -0.05, 0.05, 0.0995 and 0.13, so the upper bound holds there without conditions; for instance at $\kappa = 0.05$ the capacity is at most $\alpha_\star(0.05)$, which lies in $[0.781073068, 0.781074776]$. Three of the conditions are verified on the whole strip $\kappa \in [-0.65, 0.19]$. At $\kappa \neq 0$ the fixed-point parameters are known only to boxes about 10^5 times wider than at zero margin, and the zero-margin certificate of the two-variable condition fails at that width; it is restored by computing the moment point of the maximizing profile from the parameters instead of rounding it, and by evaluating each difference of the log-partition function as one integrand written against a fixed reference point. Whether $\alpha_\star(\kappa)$ is also a lower bound for the capacity at $\kappa \neq 0$ is open.

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1 Introduction

Store $m = \alpha N$ random patterns in a linear classifier with binary weights: for i.i.d. standard Gaussian vectors $g_1, \dots, g_m$ in $\mathbb{R}^N$ and a margin $\kappa \in \mathbb{R}$, ask whether some $w \in \{-1, +1\}^N$ satisfies

$$\frac{\langle g_a, w \rangle}{\sqrt{N}} \;\geq\; \kappa \qquad \text{for all } a \leq m.$$

Krauth and Mézard [1] computed the replica-symmetric prediction for the largest such α and conjectured it sharp; at $\kappa = 0$ their constant is $\alpha_\star = 0.8330786\ldots$, in the margin framework of Gardner and Derrida [3]. Rigorous bounds at zero margin go back to Kim and Roche [4] and Talagrand [5], who proved that the capacity is at most $1 - \varepsilon$ for some $\varepsilon > 0$; Altschuler and Tikhomirov [12] give a complete proof of the bound 0.847 outlined by Krauth and Mézard. Three decades later the zero-margin conjecture was resolved: Ding and Sun [9] proved the

lower bound and Huang [10] the upper bound. Both proofs are conditional on numerical conditions about explicit low-dimensional variational problems, and those conditions are verified in interval arithmetic in [14], at $\kappa = 0$.

Huang's upper-bound theorem, however, is stated at every margin. For each κ it bounds the capacity by $\alpha_\star(\kappa)$, the root of the replica-symmetric free energy along its fixed-point branch, conditionally on four numerical conditions at that margin. The first asks that the fixed-point equations have a unique solution, along which the free energy changes sign. The second asks that the associated approximate-message-passing iteration be stable at that solution. The third asks that an explicit function of one real variable, which controls the concavity of the first-moment functional near its maximizer, take a negative value at some point. The fourth asks that a function $\mathcal{S}_\star$ of two real variables be nonpositive on the whole plane (Conditions 1–4 of Section 2). Stojnic's fully-lifted random duality theory [22] predicts the same replica-symmetric values at every margin by nonrigorous means. We verify all four conditions at five nonzero margins.

Main Theorem (Theorem 4)**.** *For $\kappa \in \{-0.45,\ -0.05,\ 0.05,\ 0.0995,\ 0.13\}$ the four conditions of Huang's theorem hold, and the threshold $\alpha_\star(\kappa)$ lies in the interval*

$$\begin{aligned}
&\kappa = -0.45: \ [1.557652103,\ 1.557654133], \quad \kappa = -0.05: \ [0.889408031,\ 0.889409783],\\
&\kappa = 0.05: \ [0.781073068,\ 0.781074776], \quad \kappa = 0.0995: \ [0.733478513,\ 0.733480205],\\
&\kappa = 0.13: \ [0.705932217,\ 0.705933898].
\end{aligned}$$

For α above that interval, with probability $1 - o(1)$ as $N \to \infty$ no $w \in \{-1, +1\}^N$ satisfies all $\lceil \alpha N \rceil$ constraints.

At $\kappa = 0$ the four conditions are verified in [14]. There the fixed point (q_0, ψ_0) of the replica-symmetric equations (Section 2) is known to width 10^{-9}, so the moment point $a^\star = (\psi_0(1 - q_0), q_0)$ of the maximizing profile of Condition 4 can be treated as an exact point. For $\kappa \neq 0$ no published enclosure of (q_0, ψ_0) exists. Section 4 locates (q_0, ψ_0) in boxes, and at $\kappa = 0.05$ the box used in Section 6 has width $1.9 \cdot 10^{-4}$ in ψ_0. The proof of Condition 4 works in the moment body K, the set of pairs $(\mathbb{E}[\dot{H}\Lambda], \mathbb{E}[M\Lambda])$ over functions Λ of $\dot{H}$ with $|\Lambda| \leq 1$, where $\dot{H} \sim N(0, \psi_0)$ and $M = \tanh \dot{H}$. It bounds the entropy term by convex duality through the log-partition function $\Phi(b) = \mathbb{E} \log 2\cosh(b_1 \dot{H} + b_2 M)$, at dual points that must be placed by certified sign tests; these tests need the signs of differences $\Phi(\lambda) - \Phi(\lambda') - (\lambda - \lambda') \cdot x$, $x \in K$, of order 10^{-7}, while a box of that width enters each value of Φ at about $2 \cdot 10^{-5}$. The argument of [14] therefore does not apply as written. It applies after two changes. First, $a^\star$ is never stored as a number: the identity $\nabla\Phi(1, 0) = (\psi_0(1 - q_0), q_0)$ gives it, and the optimal tilt $\sqrt{1 - q_0}$, as functions of the box. Second, each such difference is written against the point $b_0 = (1, 0)$, where its integrand is

$$\log\bigl(\cosh \Delta + M \sinh \Delta\bigr) - M\Delta, \qquad \Delta = (\lambda_1 - 1)\dot{H} + \lambda_2 M,$$

so that the width of the box enters only at second order in $|\lambda - b_0|$; at the corners of the boxes the resulting enclosures are two orders of magnitude tighter than differences of separate enclosures of Φ. Two further inputs are needed: bounds on the inverse Mills ratio (Lemma 3), which keep the second derivatives of the constraint term finite where the argument of the inverse Mills ratio is large, and the one-term formula for $\partial R/\partial q$ (Lemma 2), where $\psi = R(q, \alpha)$ is the second fixed-point equation. With them the four certificates of [14] (a sweep of K away from

$a^\star$; concavity along each ray from $a^\star$ up to a radius that depends on the direction, so on a star-shaped neighborhood of $a^\star$, the star; a second sweep between the two; and a certificate that the star lies inside K) hold at $\kappa = -0.45$, -0.05, 0.05, 0.0995 and 0.13 (Sections 5.2 and 6).

Conditions 1–3 hold on the whole strip $\kappa \in [-0.65, 0.19]$, on 1680 contiguous slabs of width $5 \cdot 10^{-4}$ (Section 4 and Theorem 5). Uniqueness of the fixed point comes from a contraction bound on each slab, and on $(0.10, 0.19]$ also from Nakajima's uniqueness theorem [13], whose premise is verified in ball arithmetic. The Hessian of Huang's first-moment functional at its maximizer is negative definite at six margins (Section 5.1).

Lemma 1 is the computation of Ding and Sun [9, Lemma 7.1 and Corollary 7.2] carried out off the fixed-point branch, and Lemma 2 is Nakajima's formula [13, eq. (18)], rewritten for R; Nakajima states it for $\kappa \geq 0$, and the computation holds for every κ.

For $\kappa < 0$, Section 8 gives the lower bound $\max\{\alpha_\star(0), \alpha_c^{\mathrm{sym}}(|\kappa|)\}$, by monotonicity and by embedding a symmetric perceptron; it lies far below $\alpha_\star(\kappa)$. The enclosures of the special functions and the quadrature rules are those of [14].

Section 2 states the replica-symmetric system, Huang's four conditions and the two identities. Section 3 describes the capacity curve in floating point. Section 4 proves Conditions 1–3 on the strip, and Section 5 treats the two-variable function and its Hessian. Section 6 proves Condition 4 at the five margins and states Theorems 4 and 5; Section 7 records the further margins and the passage from points to intervals. Section 8 concerns the lower bound, Section 9 the trusted base of the computation, Section 10 the reading of the curve for learning, and Section 11 further directions. The notation is collected at the end.

2 The system and the four conditions

Let φ and Ψ denote the standard Gaussian density and upper tail, $E = \varphi/\Psi$ the inverse Mills ratio (an increasing function; $E' = E\,(E - u) \in (0, 1)$, proved with its second-derivative bounds in Lemma 3), and $Z \sim N(0, 1)$. Fix $\kappa \in \mathbb{R}$ and write

$$u(z; q, \kappa) = \frac{\kappa - \sqrt{q}\, z}{\sqrt{1 - q}}, \qquad 0 < q < 1.$$

Following Ding–Sun and Huang, the replica-symmetric system is

$$P(\psi) = \mathbb{E}\big[\tanh^2(\sqrt{\psi}\, Z)\big], \qquad R(q, \alpha) = \frac{\alpha}{1 - q}\, \mathbb{E}\big[E(u(Z))^2\big], \tag{1}$$

$$G(\alpha, q, \psi) = -\frac{\psi(1 - q)}{2} + \mathbb{E} \log\big(2 \cosh(\sqrt{\psi}\, Z)\big) + \alpha\, \mathbb{E} \log \Psi\big(u(Z)\big), \tag{2}$$

with the fixed-point branch $q = P(\psi)$, $\psi = R(q, \alpha)$ and the threshold $\alpha_\star(\kappa)$ defined as the zero of $G_\star(\alpha) = G(\alpha, q_\star(\alpha), \psi_\star(\alpha))$; we write $(q_0, \psi_0) = (q_\star, \psi_\star)(\alpha_\star(\kappa))$. At $\kappa = 0$, where $u = -\gamma z$ in law with $\gamma = \sqrt{q/(1 - q)}$, this is the zero-margin system of [14]. (Display (14) of [10] has $-\alpha\, \mathbb{E} \log \Psi$; the fixed-point equations require the plus sign, which is the sign in [9, eq. (4)].)

Huang's theorem [10, Theorem 3.6] bounds the capacity at margin κ by $\alpha_\star(\kappa)$ subject to four numerical conditions, which we state in the form his proof uses; Conditions 1, 2, 3 and 4 below are his Conditions 3.1, 3.3, 3.4 and 1.3, the conditions of his Theorem 3.6.

Condition 1 (fixed point)**.** *There are an interval* $[\alpha_-, \alpha_+]$ *and a rectangle* $[\alpha_-, \alpha_+] \times [q_{\mathrm{lb}}, q_{\mathrm{ub}}]$ *such that for every* $\alpha \in [\alpha_-, \alpha_+]$ *the map* $q \mapsto P(R(q, \alpha))$ *has a unique fixed point in* $(q_{\mathrm{lb}}, q_{\mathrm{ub}})$*, with* $\sup_{q \in (q_{\mathrm{lb}}, q_{\mathrm{ub}})} (P \circ R_\alpha)'(q) < 1$*, and* $G_\star(\alpha) = G(\alpha, q_\star(\alpha), \psi_\star(\alpha))$*, with* $q_\star(\alpha)$ *that fixed point and* $\psi_\star(\alpha) = R(q_\star(\alpha), \alpha)$*, is strictly decreasing on* $[\alpha_-, \alpha_+]$ *with a unique root there. Huang's* $\kappa = 0$ *instance (his Proposition 3.2) takes an interval of width* 10^{-9} *about the fixed point. We call the uniqueness and contraction part the rectangle clause.*

Condition 2 (approximate message passing, AMP)**.** *At the fixed point,*

$$\alpha \, \mathbb{E}[\operatorname{sech}^4(\sqrt{\psi_0} Z)] \, \frac{\mathbb{E}[E'(u(Z; q_0, \kappa))^2]}{(1 - q_0)^2} \; < \; 1,$$

with $u(z; q, \kappa) = (\kappa - \sqrt{q}\, z)/\sqrt{1 - q}$ *as above.*

Condition 3 (local concavity)**.** *Let* $A = 1 - q_0$*,* $F(x) = E\big((\kappa - x)/\sqrt{A}\big)/\sqrt{A}$*,* $\hat{f}_0 = -F'/(1 + AF')$*,* $\mu(\hat{z}) = \mathbb{E}[(\hat{z} + \cosh^2(\sqrt{\psi_0} Z))^{-1}]$*,* $d_0 = \alpha_\star \mathbb{E}[F'(\sqrt{q_0} Z)]$ *and*

$$\ell(\hat{z}) = \hat{z} - \alpha_\star \, \mathbb{E}\left[\frac{\hat{f}_0(\sqrt{q_0} Z)}{1 + \mu(\hat{z}) \hat{f}_0(\sqrt{q_0} Z)} \right] - d_0, \qquad \hat{z} > -1.$$

Then $\ell(\hat{z}) < 0$ *for some* $\hat{z} > -1$*.*

This is [10, Condition 3.4], whose function is called λ there; a single witness suffices, since Huang's condition asks that $\inf_{\hat{z} > -1} \ell(\hat{z}) < 0$.

Condition 4 (two-variable function)**.** $\mathcal{S}_\star(\lambda_1, \lambda_2) \le 0$ *for all* $(\lambda_1, \lambda_2) \in \mathbb{R}^2$*, where* $\mathcal{S}_\star$ *is the two-variable function of Section 5.2; equality holds at* $(1, 0)$*.*

2.1 Two identities

The function G is that of Ding and Sun [9, eq. (4)], who define it at general margin; the two lemmas below record its partial derivatives and the derivative of R at every κ.

Lemma 1 (stationarity)**.** *For all* $\alpha > 0$*,* $\kappa \in \mathbb{R}$*,* $\psi > 0$ *and* $0 < q < 1$*,*

$$\frac{\partial G}{\partial \psi} = \frac{q - P(\psi)}{2}, \qquad \frac{\partial G}{\partial q} = \frac{\psi - R(q, \alpha)}{2}.$$

In particular the critical points of G are exactly the solutions of Ding and Sun's fixed-point system $q = P(\psi)$, $\psi = R(q, \alpha)$ [9], at every margin.

Proof of Lemma 1. The first identity is Gaussian integration by parts on the $\log \cosh$ term. For the second, differentiating u in q gives

$$u_q = \frac{1}{2\sqrt{1 - q}} \Big(-\frac{z}{\sqrt{q}} + \frac{u}{\sqrt{1 - q}} \Big), \qquad \frac{\partial}{\partial q} \, \mathbb{E} \log \Psi(u) = -\mathbb{E}\big[E(u) \, u_q \big].$$

Stein's lemma applied to the z part, $\mathbb{E}[z E(u)] = -\sqrt{q/(1 - q)} \; \mathbb{E}[E'(u)]$, together with the Mills identity $E'(u) = E(u)^2 - u E(u)$, collapses the two pieces:

$$\mathbb{E}[E(u) \, u_q] = \frac{\mathbb{E}[E'(u)] + \mathbb{E}[u \, E(u)]}{2(1 - q)} = \frac{\mathbb{E}[E(u)^2]}{2(1 - q)} = \frac{R(q, \alpha)}{2\alpha}.$$

Hence $\partial_q G = \psi/2 - \alpha \cdot R/(2\alpha) = (\psi - R)/2$. □

Both derivatives vanish exactly on the fixed-point branch. On a rectangle carrying the contraction of Condition 1 the branch $\alpha \mapsto (q_\star, \psi_\star)$ is C^1 (the implicit function theorem applies: $1 - (P \circ R_\alpha)'$ is bounded away from zero), and the envelope theorem gives $G'_\star(\alpha) = \mathbb{E} \log \Psi(u) < 0$ strictly ($\Psi < 1$ pointwise), for every κ: the root $\alpha_\star(\kappa)$ is unique on the branch. This is the general-margin form of the monotonicity clause in Condition 1, and it is also what turns the located fixed-point boxes of Section 4 into certified $\alpha_\star$ intervals: a certified sign of G at a located box is a certified side of $\alpha_\star$. The branch is increasing in α: P is increasing, since $\tanh^2(\sqrt{\psi}\,|z|)$ increases with ψ for each z; R is linear in α and increasing in q (Lemma 2); so $q'_\star(\alpha) = P'(\psi_\star)\,(R/\alpha)\big/\big(1 - (P \circ R_\alpha)'(q_\star)\big) > 0$, and $\psi_\star = R(q_\star, \alpha)$ increases with it. Hence the hull of the boxes located at the two ends of an α interval contains $(q_\star(\alpha), \psi_\star(\alpha))$ for every α in the interval.

Lemma 2 (derivative of R)**.** *For all $\alpha > 0$, $\kappa \in \mathbb{R}$ and $0 < q < 1$,*

$$\frac{\partial R}{\partial q} = \frac{\alpha}{(1-q)^2}\,\mathbb{E}\Big[E'(u)^2 + 2\,E(u)^2 E'(u)\Big] \;>\; 0.$$

Proof. Differentiating (1) in q produces $R/(1-q) + \frac{\alpha}{1-q}\mathbb{E}[2EE'u_q]$. Splitting u_q as in Lemma 1 and applying Stein's lemma to $\mathbb{E}[z\,EE'(u)]$ (with $(EE')' = E'^2 + EE''$ and $E'' = E'(E-u) + EE' - E$) turns the cross term into

$$\mathbb{E}[2EE'u_q] = \frac{\mathbb{E}[E'^2 + 2E^2E' - E^2]}{1-q},$$

and the $-\alpha\mathbb{E}[E^2]/(1-q)^2$ piece cancels $R/(1-q)$ exactly. Positivity is immediate since $E' \in (0,1)$. In both proofs differentiation under the integral sign is dominated ($|z\tanh(\sqrt{\psi}z)| \le \sqrt{\psi}\,z^2$, and the q-integrands grow at most quadratically in $|z|$, locally uniformly in the parameters), and Stein's lemma applies because $E(u(z))$ and $(EE')(u(z))$ are absolutely continuous with bounded derivative in z. □

With $B(q) = (1-q)\mathbb{E}[E(u)^2]$ one has $R = \alpha B(q)/(1-q)^2$, and Nakajima's computation of B' [13, eq. (18)], stated for $\kappa \ge 0$, does not use the sign of κ. It gives at every margin the monotonicity of R that Ding and Sun prove at $\kappa = 0$ [9, Lemma 7.3]. The one-term form is used twice: the integrand is a product of quantities with classical bounds, so its interval enclosure is tight enough for the contraction certificate on each slab; and positivity makes $q \mapsto P(R(q))$ increasing, which is what the location step of Section 4 bisects on.

3 The capacity curve and the condition map

A warm-started continuation in high-precision floating point (not a certificate) follows the fixed point along $\kappa \in [-2.0, 1.5]$, recording at each of 71 points the threshold, the fixed point, the contraction $(P \circ R_\alpha)'(q_\star)$, the AMP product and the local-concavity minimum ℓ_0. Selected values:

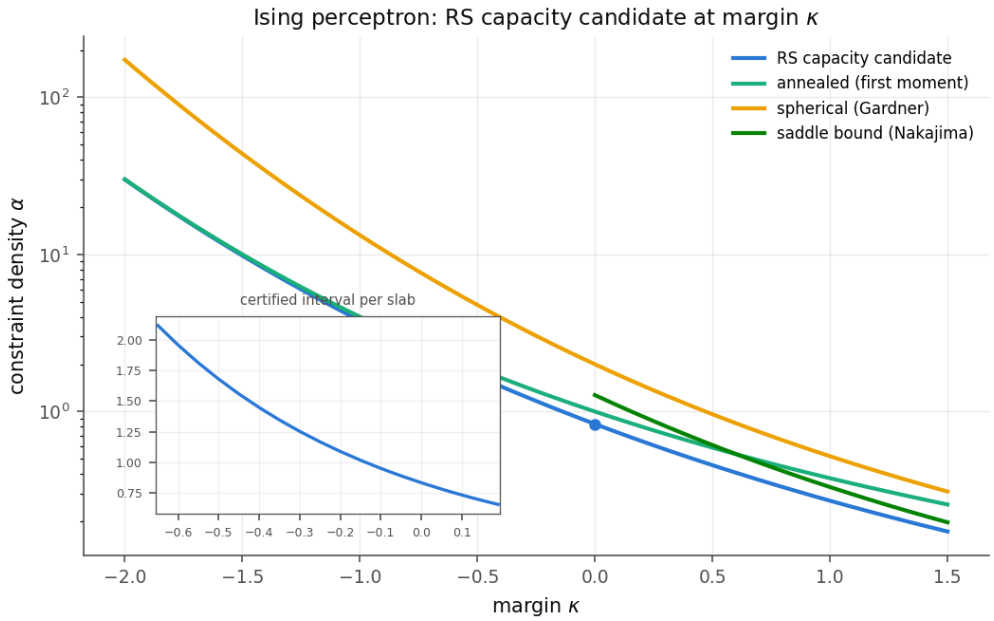


Figure 1: The replica-symmetric threshold $\alpha_\star(\kappa)$ (warm-started continuation from the certified value at $\kappa = 0$), the annealed bound $\log 2/(-\log \Psi(\kappa))$, the Gardner spherical capacity [2], and Nakajima's saddle-existence bound $\alpha_c(\kappa) = 2/(\pi\, \mathbb{E}[(\kappa - Z)_+^2])$ for $\kappa \geq 0$. The inset shows the certified per-slab $\alpha_\star$ intervals on the strip.

κ	$\alpha_\star$	$q_\star$	$(P \circ R)'$	AMP	ℓ_0
-2.0	29.9921	0.11863	0.387	0.440	-0.147
-1.0	3.79463	0.36519	0.583	0.607	-0.099
-0.5	1.67884	0.47377	0.666	0.589	-0.132
0	0.83308	0.56395	0.744	0.544	-0.191
0.5	0.45566	0.63841	0.810	0.489	-0.271
1.0	0.27070	0.69988	0.863	0.433	-0.359
1.5	0.17245	0.75049	0.903	0.381	-0.442

Conditions 1–3 hold at every point computed. The contraction is the only quantity trending toward failure, and only slowly; ℓ_0 approaches zero nearest $\kappa \approx -1.3$ (about -0.094) and then strengthens again. On the positive branch $\alpha_\star(\kappa)$ stays below Nakajima's bound at every grid point (worst ratio 0.874 at $\kappa = 1.5$). Toward $\kappa = -2$, the lower edge of the computed range, the threshold approaches the annealed bound from below, as it must where the constraints trivialize.

4 Certificates on a strip

For κ in sub-slabs of width $5 \cdot 10^{-4}$ covering $[-0.65, 0.19]$, with per-slab parameter rectangles seeded by the continuation, we verify Conditions 1–3 in ball arithmetic (`python-flint` 0.9.0 on FLINT 3.6.0, 80-bit working precision, every input a ball covering the whole slab).

4.1 Numerical architecture

The computation rests on three rules.

First, adaptive complex quadrature cannot resolve enclosure width that comes from parameter balls: subdividing the integration variable does not shrink it, and the integrator exhausts any budget. So every wide-parameter integral is a real-cell Riemann sum with exact per-cell Gaussian masses. Second, the ball evaluation of φ/Ψ fails on wide arguments in the steep region (the enclosure of Ψ crosses zero), while the ratio is monotone; all Mills-family factors are therefore enclosed by two thin endpoint evaluations, and the sign-critical sums use the mean-value cell rule (midpoint value plus a derivative correction; on a cell of width h its error is $O(h^2)$ against the plain hull's $O(h)$, with constants set by the derivative enclosures on that cell). Third, expectations that nearly cancel must be evaluated as one correlated integrand, never as separate sums whose parameter widths add. For Condition 3 this is the identity (with $\tau = E'$, $A = 1 - q_0$, and $\beta = \mu(\hat{z}) - A$ for $\mu(\hat{z}) = \mathbb{E}[(\hat{z} + \cosh^2(\sqrt{\psi_0}Z))^{-1}]$)

$$\ell(\hat{z}) = \hat{z} + \alpha\,\mathbb{E}\left[\frac{\beta\,\tau^2}{A(A+\beta\tau)}\right],$$

Indeed $F'(\sqrt{q_0}Z) = -\tau/A$, so $\hat{f}_0 = \tau/(A(1-\tau))$, $\hat{f}_0/(1+\mu\hat{f}_0) = \tau/(A+\beta\tau)$ and $d_0 = -\alpha\mathbb{E}[\tau]/A$, and $\tau/A - \tau/(A+\beta\tau) = \beta\tau^2/(A(A+\beta\tau))$. The integrand is monotone in τ for fixed (μ, A), regular on all of $\tau \in [0,1]$, and bounded by $|\beta|/(A\mu)$: the map $\tau \mapsto \tau^2/(A+\beta\tau)$ has derivative $\tau(2A+\beta\tau)/(A+\beta\tau)^2$, and $A+\beta\tau$ lies between A and $A+\beta = \mu$, both positive, so $2A + \beta\tau = A + (A+\beta\tau) > 0$; the integrand is therefore monotone in τ, and its absolute value is at most its value at $\tau = 1$, which is $|\beta|/(A\mu)$.

4.2 Uniqueness by contraction

On each slab the checks are: (i) the contraction $\sup_q (P \circ R_\alpha)' \le \sup_\psi P' \cdot \sup_q |\partial R/\partial q| < 1$ on the slab rectangle, via Lemma 2; (ii) existence: the endpoints of the q interval map strictly inward under $P \circ R_\alpha$ on each of eight α sub-intervals; (iii) the crossing: at the two thin α endpoints a certified Picard iteration locates the fixed point (the invariant is that the image of any set containing the fixed point contains it) and the sign of G resolves through a mean-value form that keeps the stationarity cancellations of Lemma 1 explicit, $G(\alpha_{\mathrm{lb}}) > 0 > G(\alpha_{\mathrm{ub}})$; (iv) Condition 2; (v) Condition 3 at Huang's witness $\hat{z} = -0.6693$. Together with the strict decrease of $G_\star$, (i)–(iii) discharge Condition 1 verbatim on the slab: $\alpha_\star(\kappa)$ exists, is unique, and lies in the slab's α interval for every κ in the slab.

The contraction bound (i) holds on each slab; over a union of slabs the enclosure of $\partial R/\partial q$ is about twice too wide for it. On a single slab the enclosure stays below 1 with slack 0.049 at $\kappa = 0$, growing to 0.198 at $\kappa = -0.65$ and thinning to 0.0083 at $\kappa = 0.10$.

The 1500 slabs are contiguous, cover $[-0.65, 0.10]$, and each satisfies (i)–(iii). The q interval of a slab must cover the drift of the fixed point across the slab's (α, κ) rectangle, which is dominated by $\partial q_\star/\partial\alpha \approx 0.92$ times the width of the α interval; the drift $dq_\star/d\kappa$ stays near 0.18 across the strip.

4.3 Uniqueness on $(0.10, 0.19]$

The product of suprema exceeds 1 near $\kappa = 0.12$, while the contraction at the fixed point stays below 0.78: at $\kappa = 0.13$ the product over the slab rectangle is 1.32, while a certified sign

bisection of the displacement $P(R(q)) - q$ (which has certified signs at the interval endpoints and, by the uniqueness of the fixed point, a single zero between them, so it is positive below the fixed point and negative above) locates the fixed point at both α endpoints to width $2 \cdot 10^{-8}$ with no contraction input, the contraction evaluated on the located intervals reads 0.79, and the G crossing resolves with slack $2 \cdot 10^{-3}$.

The rectangle clause itself is then discharged at $\kappa = 0.13$ exactly as Huang's Proposition 3.2 discharges it at $\kappa = 0$: on the rectangle $\alpha \in [0.705932217, 0.705933898]$, $q \in [0.5836943, 0.5857104]$ (the located fixed points $\pm 10^{-3}$), the certified contraction product is 0.8435, the map is endpoint-inward at both α ends with slack above $2.3 \cdot 10^{-4}$, the G signs at the located fixed points are $+1.6 \cdot 10^{-7}$ and $-1.6 \cdot 10^{-6}$, and $\mathbb{E} \log \Psi < 0$ over the rectangle (`rect_cond31_0p13.json`; quadrature depth 16000). The value 1.32 comes from the width of the slab rectangle; the contraction at the fixed point $(P \circ R_\alpha)'(q_\star)$ is nearly flat, rising only from 0.758 at $\kappa = 0.10$ to 0.770 at $\kappa = 0.19$, everywhere below 1 by more than 0.22. The rectangle clause is therefore decided by the width of the rectangle: over $(0.10, 0.19]$ (`rect_strip_nak.jsonl`, per κ sub-ball) the product is below 1 at $\kappa = 0.10$ with q half-width $2 \cdot 10^{-3}$ but exceeds it (1.012) by $\kappa = 0.15$ with the same half-width, and a thinner located rectangle brings $\kappa = 0.13$ to 0.844. The half-width is set by the located q interval, whose width comes mostly from the slab's α width ($3 \cdot 10^{-3}$) through $\partial q_\star / \partial \alpha \approx 0.9$; narrowing it means deep-locating $\alpha_\star$ per slab to $\sim 10^{-6}$, the same tightening the five point margins already carry, after which the flat contraction leaves a thin rectangle that clears 1. At $\kappa \in \{0.13, 0.15, 0.185\}$ the contraction products on such rectangles are 0.844, 0.836 and 0.830 (`rect_cond31_0p15.json`, `rect_cond31_0p185.json`). On each of the 180 slabs covering $[0.10, 0.19]$, over eight κ sub-balls of width $6.25 \cdot 10^{-5}$, the located rectangle with q half-width $5 \cdot 10^{-4}$ is endpoint-inward at both α ends and carries a certified contraction product below one, the worst bound being 0.958 (`rect_strip_nak_thin_*.jsonl`, collected with the input hashes bound in `rect_strip_thin_summary.json`). With the inward endpoints and Lemma 2's monotonicity this gives, for every $\kappa \in (0.10, 0.19]$, a rectangle on which the fixed point is unique by the contraction itself, so the rectangle clause of Condition 1 holds on every slab with uniqueness carried both by the rectangle and, independently, by Nakajima's theorem.

The other checks on these 180 slabs are: existence by the same inward test; location by certified sign bisection per κ sub-ball (the κ width floors the displacement's sign resolution, so each slab runs eight sub-balls whose union covers it); the contraction used only on located intervals, which is the local form Huang's perturbation argument uses; and uniqueness from Nakajima's saddle theorem [13], whose premise $\alpha_{\mathrm{ub}} < \alpha_c(\kappa) = 2/(\pi\, \mathbb{E}[(\kappa - Z)_+^2])$ is certified per slab through the closed form $\mathbb{E}[(\kappa - Z)_+^2] = (1 + \kappa^2)\big(1 - \Psi(\kappa)\big) + \kappa\varphi(\kappa)$ (least certified premise slack 0.311).

On the whole strip, 1680 contiguous slabs pass and none fails. At five margins a certified bisection on the sign of G at located boxes tightens the slab intervals to width about $2 \cdot 10^{-6}$; the resulting enclosures are those of Theorem 4. Each contains its continuation value, and the slab containing $\kappa = 0$ contains the Ding–Sun constant. At the default quadrature depth the enclosure of G limits the width to $1.4 \cdot 10^{-5}$; the five intervals use a larger depth.

5 The two-variable function

This section treats the first-moment functional at its maximizer and writes Huang's two-variable function at general margin in the form the certificates of Section 6 evaluate.

5.1 The maximizer and its Hessian

The local ingredient of Condition 4 is the negative definiteness of the Hessian $\mathcal{H}$ of Huang's first-moment functional at its maximizer. His second-derivative expansion, *before* the zero-margin specialization, gives at every κ the same quadratic form

$$\langle \xi, \mathcal{H}\xi \rangle = -(I_1\xi_1^2 + 2I_2\xi_1\xi_2 + I_3\xi_2^2) + C_1 y_1^2 + C_2 y_1 y_2 + C_3 y_2^2,$$

where $y_1 = I_1\xi_1 + I_2\xi_2$ and $y_2 = I_2\xi_1 + I_3\xi_2$, with prior-side integrals I_j unchanged in form ($I_1 = \psi_0(1-q_0) - 2\psi_0^2(1-4q_0+3p_4)$, $I_2 = \psi_0(1-4q_0+3p_4)$, $I_3 = q_0 - p_4$, $p_4 = \mathbb{E}\tanh^4(\sqrt{\psi_0}Z)$) and constraint-side constants C_j that at $\kappa \neq 0$ must be taken from the pre-specialization display. Put $\hat{H} = \sqrt{q_0}\,Z$, $u = (\kappa - \hat{H})/\sqrt{1-q_0}$, $\eta = E(u)/\sqrt{1-q_0}$ and $F' = -E'(u)/(1-q_0)$. His zero-margin closed forms rest on two identities of the unshifted law, of which $\alpha\,\mathbb{E}[\eta^2] = \psi_0$ survives at every margin (it is the same identity behind the unbounded-tilt directions of Section 5.2) while $\alpha\,\mathbb{E}[\eta\hat{H}] = -q_0(1-q_0)\psi_0$ does not. We evaluate the general-margin C_j directly. With $Q = -\hat{H}/q_0 + \big(\kappa - \hat{H} - (1-q_0)\eta\big)/(1-q_0)$, every expectation over the single Gaussian $\hat{H}$,

$$C_1 = \frac{\alpha}{\psi_0^2}\,\mathbb{E}[F'\eta^2], \qquad C_2 = -\frac{2\alpha}{\psi_0}\,\mathbb{E}[F'\eta Q] + \frac{2}{1-q_0},$$

$$C_3 = \alpha\,\mathbb{E}[F'Q^2] + \frac{2\alpha}{q_0(1-q_0)}\,\mathbb{E}[\eta\hat{H}] - \alpha\Big(\frac{3}{(1-q_0)^2} + \frac{1}{q_0(1-q_0)}\Big)\mathbb{E}\big[\eta\,(\kappa - \hat{H} - (1-q_0)\eta)\big].$$

The certified evaluation encloses each expectation by the mean-value cell rule with explicit quadratic tail majorants (η grows at most linearly on the left tail, F' is bounded by $1/(1-q_0)$ in magnitude, and every integrand is degree at most two in $|z|$), and the near-cancelling prior moments are evaluated as single correlated integrands. The certified enclosures at the six margins are one-sided where sign is the claim:

$$\begin{array}{ll}
\kappa = -0.45: & \mathcal{H}_{11} \le -0.0647, \quad \det\mathcal{H} \ge 2.9\cdot 10^{-4},\\
\kappa = -0.05: & \mathcal{H}_{11} \le -0.0471, \quad \det\mathcal{H} \ge 1.6\cdot 10^{-4},\\
\kappa = 0: & \mathcal{H}_{11} \le -0.0451, \quad \det\mathcal{H} \ge 1.8\cdot 10^{-4},\\
\kappa = 0.05: & \mathcal{H}_{11} \le -0.0424, \quad \det\mathcal{H} \ge 1.2\cdot 10^{-4},\\
\kappa = 0.0995: & \mathcal{H}_{11} \le -0.0400, \quad \det\mathcal{H} \ge 1.0\cdot 10^{-4},\\
\kappa = 0.13: & \mathcal{H}_{11} \le -0.0385, \quad \det\mathcal{H} \ge 8.8\cdot 10^{-5}.
\end{array}$$

The bound on $|\mathcal{H}_{11}|$ decreases as κ increases.

Certification requires parameter boxes far tighter than the slabs provide (the C_j move with q at rate about 10, while $\det\mathcal{H} \sim 10^{-4}$), which is what the deep $\alpha_\star$ intervals above are for. Iterating $q \mapsto P(R(q,\alpha))$ over the whole deep interval gives boxes that contain the fixed point for every α in it ($3.6\cdot 10^{-6}$ in q_0, $5.2\cdot 10^{-5}$ in ψ_0 at $\kappa = 0.05$). On these boxes the certificate resolves: $\mathcal{H}_{11} < 0$ and $\det\mathcal{H} > 0$ in ball arithmetic, with the near-cancellations $1 - 4q_0 + 3p_4$ and $q_0 - p_4$ evaluated as single correlated integrands and the five constraint-side integrals by mean-value Riemann sums with elementary Mills-family derivatives. The same chain certifies at $\kappa = 0$ on the Ding–Sun rectangle, where the C_j enclosures lie inside Huang's certified intervals, and at -0.05, 0.0995, -0.45 (where $C_1 = -0.855$, against -0.718 at zero margin) and 0.13.

Along the continuation the fixed-tilt $\mathcal{H}$ stays negative definite at least up to $\kappa = 1.15$ and loses definiteness inside $(1.1812, 1.1844)$, while the Hessian of the tilt-optimized functional

remains negative out to at least $\kappa = 1.5$; the two are consistent by the Schur relation (the fixed-tilt form dominates), so past $\kappa \approx 1.18$ a local argument must re-optimize the tilt, as the ray certificates below already do.

5.2 The function at general margin

Huang's two-variable function at margin κ, in the form his proof uses, is

$$\mathcal{S}(\Lambda, s) = \frac{s^2\psi_0}{2} + \mathrm{ent}(\Lambda) + \alpha\,\mathbb{E}\log\Psi\left(\frac{\kappa - (a_2/q_0)\hat{H} - (a_1/\psi_0)\eta}{D} + s\eta\right),$$

where $\dot{H} \sim N(0,\psi_0)$ and $\hat{H} \sim N(0,q_0)$ are independent, $M = \tanh\dot{H}$ (Huang's notation), the profile Λ is a function of $\dot{H}$ with $|\Lambda| \le 1$, $\mathrm{ent}(\Lambda) = \mathbb{E}\,h\big(\frac{1+\Lambda}{2}\big)$ with $h(p) = -p\log p - (1-p)\log(1-p)$, $\eta = E(u)/\sqrt{1-q_0}$ is the variable of Section 5.1 (the margin enters through η and through the additive κ in the $\log\Psi$ argument), $a_1 = \mathbb{E}[\dot{H}\Lambda]$, $a_2 = \mathbb{E}[M\Lambda]$, $D = \sqrt{1-a_2^2/q_0}$, and $\mathcal{S}_\star(\lambda_1,\lambda_2) = \inf_{s\ge 0}\mathcal{S}(\tanh(\lambda_1\dot{H} + \lambda_2 M), s)$. Condition 4 asks $\mathcal{S}_\star \le 0$ everywhere.

The reduction of [14] holds at every κ: the profile enters only through the moment pair (a_1, a_2), which ranges over a compact convex body K (the moment body of $(\dot{H}, M)$), and on K the entropy term is bounded through $H(a) = \max\{\mathrm{ent}(\Lambda) : \mathbb{E}[\dot{H}\Lambda] = a_1,\ \mathbb{E}[M\Lambda] = a_2\}$ by convex duality, $H(a) \le \Phi(b) - b\cdot a$ for every dual point b, with $\Phi(b) = \mathbb{E}\log 2\cosh(b_1\dot{H} + b_2 M)$ free of κ. The margin lives entirely in the constraint term. Put $D(a) = \sqrt{1-a_2^2/q_0}$,

$$\mathcal{F}(a,s) = \frac{s^2\psi_0}{2} + H(a) + \alpha\,\mathbb{E}\log\Psi\left(\frac{\kappa - (a_2/q_0)\hat{H} - (a_1/\psi_0)\eta}{D(a)} + s\eta\right), \qquad \bar{\mathcal{S}}(a) = \inf_{s\ge 0}\mathcal{F}(a,s).$$

Since $\mathrm{ent}(\Lambda) \le H(a(\Lambda))$, with $a(\Lambda) = (\mathbb{E}[\dot{H}\Lambda], \mathbb{E}[M\Lambda])$, and the constraint term depends on Λ only through $a(\Lambda)$, we have $\mathcal{S}(\Lambda, s) \le \mathcal{F}(a(\Lambda), s)$, hence

$$\mathcal{S}_\star(\lambda) \le \bar{\mathcal{S}}\big(a(\tanh(\lambda_1\dot{H} + \lambda_2 M))\big), \qquad \lambda \in \mathbb{R}^2.$$

The certificates bound $\mathcal{F}$ with $H(a)$ replaced by $\Phi(b) - b\cdot a$ at the dual point b of each cell. Two structural facts organize the verification. The maximizing profile $\tanh\dot{H}$, that is $\lambda = (1,0)$, has moment point $a^\star = (\psi_0(1-q_0),\, q_0)$ for every κ: this is the base-point identity $\nabla\Phi(1,0) = (\mathbb{E}[\dot{H}M], \mathbb{E}[M^2]) = (\psi_0(1-q_0), q_0)$ (Gaussian integration by parts and the fixed-point equations), and [10, Lemma 7.15] gives $\mathcal{S}_\star = 0$ there with optimal tilt $s_0 = \sqrt{1-q_0}$, both in closed form. And on a region close to the half-plane $\lambda_1 < 0$ the s-infimum is $-\infty$: $\alpha\,\mathbb{E}[\eta^2] = \psi_0$ exactly at the fixed point, so the quadratic tilt coefficient cancels, and wherever the surviving linear coefficient of the tilt is nonpositive the functional runs away (the measured boundary of that region at $\kappa = 0.05$ crosses $\lambda_1 = 0.04$ on the $\lambda_2 = 0$ axis and bends left as $|\lambda_2|$ grows). Condition 4 takes the infimum over all $s \ge 0$, so a negative value at one finite tilt is a witness, and the Region II certificates exhibit one cell by cell.

In floating point the function is nearly the same at every κ of the curve: a single peak at $(1,0)$, off-center values below $-3\cdot 10^{-3}$, negative far rays, Hessian eigenvalues near -0.22 and -0.0045, distance near 0.017 from the boundary of K, with the weak eigenvalue degrading toward deep negative margin (-0.0013 at $\kappa = -1.5$).

6 Certifying the two-variable condition

The zero-margin certificate has four parts: a sweep of the moment body outside an exclusion box around $a^\star$ (Region II), banded ray-concavity certificates on a star around $a^\star$ (Region I), a second sweep of the exclusion box minus the star (stage 2), and an interior-ball certificate placing the star strictly inside the moment body. At $\kappa \neq 0$ all four meet the same difficulty. At zero margin the parameters (q_0, ψ_0) are certified to a rectangle of width 10^{-9}, so the maximizer's coordinates can be stored as decimals and every enclosure treats them as exact; at $\kappa = 0.05$ the hull of the boxes located at the two ends of the deep interval has width $1.9 \cdot 10^{-6}$ in q_0 and $1.9 \cdot 10^{-4}$ in ψ_0, and that width enters every plain evaluation of Φ at about $2 \cdot 10^{-5}$, much wider than the differences of order 10^{-7} whose signs place the dual points (the dual localization below). Three changes make the four certificates apply at $\kappa \neq 0$: $a^\star$ computed from the box, differences of Φ written against $b_0 = (1, 0)$, and kernels that stay finite where the argument of the inverse Mills ratio is large.

The moment point from the box. Every certificate that references $a^\star$ or the base tilt s_0 uses them as the functions $(\,\psi_0(1-q_0),\, q_0\,)$ and $\sqrt{1-q_0}$ of the box, never as stored decimals, so the true $a^\star$, at the true parameters, is covered exactly and the ray identities ($\mathcal{S} = 0$, $\nabla\mathcal{S} = 0$, tilt optimality at the true $a^\star$) hold on the certified rays with no enlargement of the box of $a^\star$. Certificates evaluated over the box of $a^\star$ cover in particular the rays from the true $a^\star$.

The ray argument. For a unit direction v and a slope $c \in \mathbb{R}$ put $\chi_{v,c}(t) = \mathcal{F}(a^\star + tv,\, s_0 + ct)$, with $\mathcal{F}$ as in Section 5, so that $\bar{\mathcal{S}}(a^\star + tv) \le \chi_{v,c}(t)$ whenever $s_0 + ct \ge 0$. At the true parameters $\chi_{v,c}(0) = \chi'_{v,c}(0) = 0$ for every c, by the value, gradient and tilt identities at $a^\star$. If $\chi''_{v,c} \le 0$ on $[0, T(v)]$ then $\chi_{v,c} \le 0$ there, hence $\bar{\mathcal{S}} \le 0$ on the segment. The chain needs one slope per ray: the second-derivative bound on a radial band $[t_0, t_1]$ with $t_0 > 0$ says nothing about χ itself unless χ and χ' are controlled at t_0, and those are controlled only through the same function on $[0, t_0]$. The certificates therefore fix the slope by direction alone: the circle is cut into 768 equal sectors, each sector carries one slope (the optimal tilt slope at its midpoint, rounded to six decimals), every angular cell of every radial band lies inside one sector and uses that sector's slope, and a direction on a sector edge is covered by the cells of either neighbor, each with its own fixed slope. Each cell lies in one sector, the cells of each band tile its angle range, and the main and supplementary certificates use the same slope table.

Differences against b_0. The dual localization (the argument that the dual maximizer $\lambda(x)$ stays inside a certified box, so that the entropy Hessian can be bounded there) needs signs of differences $\Phi(\lambda) - \Phi(\hat{\lambda}_k) - (\lambda - \hat{\lambda}_k) \cdot x$ with $x = a^\star + \text{offset}$. Evaluated as differences of plain Φ enclosures these carry the full parameter fuzz twice. Written against $b_0 = (1, 0)$, the integrand collapses exactly: with $\Delta = (\lambda_1 - 1)\dot{H} + \lambda_2 M$,

$$\log \frac{\cosh(\dot{H} + \Delta)}{\cosh \dot{H}} - (\lambda - b_0) \cdot \nabla_b \log 2\cosh(b \cdot (\dot{H}, M))\big|_{b_0} = \log\big(\cosh \Delta + M \sinh \Delta\big) - M\Delta,$$

because the gradient of $b \mapsto \log 2\cosh(b \cdot (\dot{H}, M))$ at b_0 is $M\,(\dot{H}, M)$, whose inner product with $\lambda - b_0$ is $M\Delta$. Every factor is a small ball with thin coefficients, so the parameter width enters only at second order in $|\lambda - b_0|$; at the box corners, where plain differences of enclosures fail, the enclosures are two orders of magnitude tighter, and at b_0 the evaluator returns the exact zero ball. Differences between arbitrary points are then differences of two such enclosures, and the gradient term the localization needs is the same trick applied to $\nabla\Phi$: $\nabla\Phi(\lambda) - a^\star$ has the exact per-node form $(\dot{H}, M)\,(1 - M^2)\tanh\Delta/(1 + M\tanh\Delta)$.

The factor $\cosh\Delta + M\sinh\Delta$ itself has two exact evaluations with disjoint stability regimes: the cosh/sinh form is exact at $\Delta = 0$ (which the inner-band gaps of order 10^{-7} require) but subtracts two $e^{|\Delta|}$-sized balls when $M \approx \pm 1$ and $|\Delta|$ is large, where the guard against the complex logarithm's branch then fires on every subdivision; the split $e^{\Delta}(1+M)/2 + e^{-\Delta}(1-M)/2$ has both terms nonnegative on the real axis but carries first-order M-width near $\Delta = 0$. The evaluator branches per node at $|\Delta| = 1$; both forms are identities, so the branch affects tightness only.

Kernels at large arguments. The ray certificates use second derivatives of the constraint term, whose kernels involve $E'(V) = E(V)(E(V) - V)$ and $E''(V)$, with E the inverse Mills ratio of Section 2, evaluated at cavity arguments V that reach about 20 on far quadrature cells; there $E(V) - V$ is a difference of two large balls and the enclosures of the kernels are useless. The bounds of the following lemma, applied as intersections with the computed balls, remove the failure; the intersection of an enclosure with a proven bound is again an enclosure.

Lemma 3 (bounds on the inverse Mills ratio)**.** *Let E be the inverse Mills ratio and $\sigma = E - V$. For every $V \in \mathbb{R}$,*

$$\sigma > 0 \quad \Big(\textit{and } \sigma < \frac{1}{V} \textit{ for } V > 0\Big), \qquad E' = E\sigma \in (0,1), \qquad 0 < E'' < \sigma.$$

Proof. For $V \le 0$, $\sigma > 0$ because $E > 0$. For $V > 0$ the bounds $V < E(V) < V + 1/V$ are Gordon's inequalities [7], which give $0 < \sigma < 1/V$. The bounds $0 < E' < 1$ are due to Sampford [8]; in the present notation $E' = E^2 - VE = E\sigma$, and $1 - E' = \mathrm{Var}(Z \mid Z > V)$ by the conditional moments $\mathbb{E}[Z \mid Z > V] = E$ and $\mathbb{E}[Z^2 \mid Z > V] = 1 + VE$, which are used again below. The upper bound on E'' is the identity $E'' - \sigma = (E' - 1)(E + \sigma)$ with $E' < 1$ and $E + \sigma > 0$.

Positivity of E'' is the convexity of the inverse Mills ratio; we include a short proof. Differentiating E' gives $E'' = E\,[\,\sigma(E+\sigma) - 1\,]$, and for W distributed as $Z - V$ conditioned on $\{Z > V\}$ the moment formulas above read $\mathbb{E}W = \sigma$ and $\mathbb{E}[W^2] = 1 - VE + V^2$, whence $\sigma(E+\sigma) - 1 = 2(\mathbb{E}W)^2 - \mathbb{E}[W^2]$: the claim is that this residual law has coefficient of variation below one. Its survival function is $\mathbb{P}(W > w) = \Psi(V+w)/\Psi(V) = e^{-\Gamma(w)}$ with $\Gamma(w) = \int_0^w E(V+x)\,dx$ convex, since $E' > 0$. Set $\theta = 1/\sigma$, so the rate-θ exponential has the same mean, and $J(w) = \Gamma(w) - \theta w$. Then J is convex with $J(0) = 0$, $J'(0) = E - 1/\sigma < 0$ (this is $E\sigma = E' < 1$ again), and $J(w) \to \infty$, so $J < 0$ exactly on some interval $(0, w_0)$ and $J > 0$ past it. Hence $\mathbb{P}(W > w) - e^{-\theta w} = e^{-\theta w}(e^{-J(w)} - 1)$ is positive before w_0 and negative after, while $\int_0^\infty (\mathbb{P}(W > w) - e^{-\theta w})\,dw = \sigma - \sigma = 0$. Therefore

$$\tfrac{1}{2}\,\mathbb{E}[W^2] - \sigma^2 = \int_0^\infty w\,\big(\mathbb{P}(W > w) - e^{-\theta w}\big)\,dw = \int_0^\infty (w - w_0)\big(\mathbb{P}(W > w) - e^{-\theta w}\big)\,dw < 0,$$

the last integrand being nonpositive and not almost everywhere zero. So $\mathbb{E}[W^2] < 2(\mathbb{E}W)^2$, that is $E'' > 0$, at every real V. □

With these three changes the four certificates hold at each of the five margins. We give the counts at $\kappa = 0.05$; the other margins are in Section 7. At $\kappa = 0.05$, on that hull and the deep $\alpha_\star$ interval of Section 4:

(i) *Region II*: 1296 top cells over the moment rectangle minus the exclusion box, refining to 855 leaves, zero failures.

(ii) *Region I*: the banded ray-concavity certificates over the star (radius 0.012 along the weak direction), 1404 bands from $a^\star$, zero failures; the certified band values lie between $-1.6 \cdot 10^2$ and $-7.5 \cdot 10^{-5}$. A supplementary run over twelve further angular ranges extends the certified radius to 0.0105 on the arcs 0.14 to 0.19 radians off the weak axes (certified values $-3.5 \cdot 10^{-3}$ to -1.9), which the stage-2 leaves near those arcs require.

(iii) *Stage 2*: the exclusion box minus the star, 1363 leaves, zero failures. Star containment uses the certified radius of Region I: a certified band covers every direction of its angular range out to the star boundary, so the certified radius is the pointwise maximum of the main radius and the supplementary arcs, a piecewise-constant function of the angle.

(iv) *Star interior*: the star, with the box of $a^\star$, lies strictly inside the moment body: constructive inradius 0.01481 against required 0.01209.

Together: $\mathcal{S}_\star \le 0$ on the whole plane at $\kappa = 0.05$. All four of Huang's conditions hold there, and his theorem gives the unconditional bound.

Theorem 4. *Let $\kappa \in \{-0.45, -0.05, 0.05, 0.0995, 0.13\}$. Then $\alpha_\star(\kappa)$ lies in the corresponding interval*

$$\alpha_\star(-0.45) \in [1.557652103,\ 1.557654133], \quad \alpha_\star(-0.05) \in [0.889408031,\ 0.889409783],$$
$$\alpha_\star(0.05) \in [0.781073068,\ 0.781074776], \quad \alpha_\star(0.0995) \in [0.733478513,\ 0.733480205],$$
$$\alpha_\star(0.13) \in [0.705932217,\ 0.705933898],$$

and for every α above that interval

$$\lim_{N\to\infty} \mathbb{P}\Big(\exists\, w \in \{-1,+1\}^N : \ \frac{\langle g_a, w\rangle}{\sqrt{N}} \ge \kappa \text{ for all } a \le \lceil \alpha N \rceil\Big) = 0.$$

The certificates of this section accept the located boxes of any slab. The other three conditions hold on the whole strip. The 1680 slabs, each with its κ, α and q intervals, are listed in the files `certified_intervals.csv` ($\kappa \le 0.10$) and `certified_intervals_nak.csv` ($\kappa > 0.10$) of the release.

Theorem 5. *(a) For every $\kappa \in [-0.65, 0.10]$, Conditions 1–3 hold, with the contraction of Condition 1 on the slab rectangle.*

(b) For every $\kappa \in (0.10, 0.19]$, Conditions 1–3 hold, with the contraction of Condition 1 on a located rectangle of q half-width $5 \cdot 10^{-4}$; uniqueness of the fixed point also follows from Nakajima's theorem [13], whose premise is verified on every slab.

(c) In both cases $\alpha_\star(\kappa)$ exists, is unique on the fixed-point branch, and lies in the α interval of the slab containing κ. If Condition 4 also holds at κ, then for every α above the upper end of that interval the probability that some $w \in \{-1,+1\}^N$ satisfies all $\lceil \alpha N \rceil$ constraints at margin κ tends to 0 as $N \to \infty$.

Condition 4 holds at the five margins of Theorem 4 and, by [14], at $\kappa = 0$. At every other κ of the strip, Theorem 5(c) reduces the upper bound to Condition 4 at that κ.

κ	$\alpha_\star(\kappa)$ interval	Region II	Region I	stage 2	inradius / required
0.05	[0.781073068, 0.781074776]	855	1404 (+12)	1363	0.01481 / 0.01209
−0.45	[1.557652103, 1.557654133]	656	1404 (+12)	1196	0.01480 / 0.01207
−0.05	[0.889408031, 0.889409783]	867	1404 (+12)	1282	0.01527 / 0.01209
0.0995	[0.733478513, 0.733480205]	904	1404 (+12)	1390	0.01459 / 0.01209
0.13	[0.705932217, 0.705933898]	874	1404 (+12)	1422	0.01446 / 0.01210

Table 1: Certified leaf and band counts per piece, with zero failures in every manifest. Region I counts marked (+12) include the twelve supplementary angular ranges, which the stage-2 lattice of every margin uses. The final column gives the constructive moment-body inradius against the required star radius, both exact-arithmetic bounds.

7 Further margins

Huang's four conditions are statements at one margin, and the certificates of Sections 5.2–6 run unchanged at any κ of the strip, with the boxes of (q_0, ψ_0) located at both ends of the α interval of that margin. Besides $\kappa = 0.05$, Condition 4 holds at $\kappa = -0.45$, -0.05, 0.0995 and 0.13. At the two negative margins the fixed-point uniqueness input comes entirely from the contraction certificates of Section 4. Table 1 lists the certified pieces at the five margins.

For each band, the Region-I manifest of every margin carries an interval matrix B with $\nabla^2\Phi(\lambda) \preceq B$ in the Loewner order (the difference is positive semidefinite) for every λ in the band's certified dual box, so that the Hessian of the entropy term is at most $-B^{-1}$ there. Every band's B is positive definite, a fact checked in exact rational arithmetic (Section 9).

The stage-2 containment reads each supplement arc on both angular branches, so a cell sees every arc whichever branch its interval lies on. At $\kappa = 0.0995$ the exclusion box has upper offset +0.1030; the stage-2 containment derives its tiling from the box, and the star and the supplement cover it.

7.1 From points to intervals

For Conditions 1–3 the strip certificates already give statements on intervals of κ: each slab is certified with every input enclosed over the whole slab. Condition 4 is different. Over a slab every certificate must hold with (α, q_0, ψ_0) replaced by their hulls, and with the present locating algorithm these hulls do not shrink below about $3.5 \cdot 10^{-4}$ in q_0; the induced enclosure of $a^\star$ has radius at least a few times 10^{-3} however narrow the slab, and at that radius the Region II cell enclosures are about ± 0.2, while the thinnest Region II cells have slack $5 \cdot 10^{-4}$ and the thinnest Region I bands between $1.1 \cdot 10^{-5}$ and $7.5 \cdot 10^{-5}$. An interval theorem for Condition 4 therefore needs a certified κ-derivative of each certificate, with the fixed-point branch differentiated implicitly, and quadratic control with explicit remainder bounds on the annulus between the star and the far cells, where the certified Hessian alone does not reach.

8 Toward the matching lower bound

Write $m_N(\kappa)$ for the largest m such that some $w \in \{-1, +1\}^N$ satisfies $\langle g_a, w\rangle/\sqrt{N} \geq \kappa$ for all $a \leq m$. Krauth and Mézard conjectured the value of the limiting ratio $m_N(\kappa)/N$ for every margin, not only at $\kappa = 0$ [1]; the constant 0.833 is the zero-margin point of the conjectured

curve $\alpha_\star(\kappa)$. At $\kappa = 0$ the conjecture is resolved: Ding and Sun [9] proved $\alpha_\star$ is a lower bound with positive probability, the sharp-threshold sequence of Xu [11] and Nakajima and Sun [21] upgrades positive probability to convergence in probability, Huang [10] proved the matching upper bound, and the numerical hypotheses of both proofs are certified [14]. Away from $\kappa = 0$ the upper half is Theorem 4. For the lower half, the replica-symmetric free energy is proved at small constraint density by Talagrand [6] and, for a wider class of activations, by Bolthausen, Nakajima, Sun, and Xu [15]; what has no analytic theorem to certify at present is a lower bound matching $\alpha_\star(\kappa)$. There are three partial results: a lemma showing that plain second moments fail at every margin, an embedded bound that gives capacity lower bounds for $\kappa < 0$, and a floating-point evaluation of the κ-dependent inputs of the Ding–Sun argument at the five margins.

8.1 Plain second moments fail at every margin

Write $\mathcal{Z} = \mathcal{Z}_{N,\kappa}$ for the number of $w \in \{-1,+1\}^N$ satisfying all $m = \lceil \alpha N \rceil$ constraints, and $\varpi_\rho(\kappa) = \mathbb{P}(X \ge \kappa,\, Y \ge \kappa)$ for (X, Y) bivariate standard Gaussian with correlation ρ.

Lemma 6. *For every $\kappa \in \mathbb{R}$ and $\alpha > 0$,*

$$\liminf_{N\to\infty} \frac{1}{N} \log \frac{\mathbb{E}[\mathcal{Z}^2]}{(\mathbb{E}\mathcal{Z})^2} \;\ge\; \delta(\alpha,\kappa) \;>\; 0.$$

Consequently the guarantee of the Paley–Zygmund bound, $\mathbb{P}(\mathcal{Z} > 0) \ge (\mathbb{E}\mathcal{Z})^2/\mathbb{E}[\mathcal{Z}^2]$, is at most $e^{-\delta N + o(N)}$, and no bound depending on $\mathbb{E}\mathcal{Z}$ and $\mathbb{E}[\mathcal{Z}^2]$ alone can certify more, since a distribution placing mass $(\mathbb{E}\mathcal{Z})^2/\mathbb{E}[\mathcal{Z}^2]$ at $\mathbb{E}[\mathcal{Z}^2]/\mathbb{E}\mathcal{Z}$ and the rest at zero matches both moments. For $\kappa \ge 0$ the same conclusion holds for every weighted count $\mathcal{Z}_f = \sum_w \prod_a f(\langle g_a, w\rangle/\sqrt{N})$ with $f \ge 0$ supported on $[\kappa,\infty)$ and $0 < \int f\varphi\, dx < \infty$.

Proof. $\mathbb{E}\mathcal{Z} = 2^N \Psi(\kappa)^m$ by independence over constraints. For a pair w, w' at overlap ρ the constraint fields are bivariate Gaussian with correlation ρ, so counting ordered pairs by Hamming distance k,

$$\mathbb{E}[\mathcal{Z}^2] = 2^N \sum_{k=0}^{N} \binom{N}{k} \varpi_{1-2k/N}(\kappa)^m.$$

Fix $\rho_0 \in (0,1)$, let $k^* = \lceil N(1-\rho_0)/2 \rceil$ and $\rho^* = 1 - 2k^*/N$. Keeping the single term k^* and using $\binom{N}{k} \ge e^{Nh(k/N)}/(N+1)$ with h the natural-log binary entropy ($h(p) = h(1-p)$, so the exponent may be written with $(1+\rho_0)/2$),

$$\frac{1}{N}\log\frac{\mathbb{E}[\mathcal{Z}^2]}{(\mathbb{E}\mathcal{Z})^2} \ge \underbrace{h\Big(\tfrac{1+\rho_0}{2}\Big) - \log 2 + \alpha\big[\log \varpi_{\rho_0}(\kappa) - 2\log\Psi(\kappa)\big]}_{=:\,\zeta_{\alpha,\kappa}(\rho_0)} + O(N^{-1}\log N),$$

the error, of order $N^{-1}\log N$, collecting the binomial prefactor, the $|\rho^* - \rho_0| \le 2/N$ continuity of $\rho \mapsto \log\varpi_\rho$ (differentiable, below), and $|m - \alpha N| \le 1$. Now $\varpi_0 = \Psi(\kappa)^2$ and $h(1/2) = \log 2$ give $\zeta_{\alpha,\kappa}(0) = 0$. Reflecting the orthant to a joint CDF, $\varpi_\rho(\kappa) = N_2(-\kappa,-\kappa;\rho)$ with N_2 the bivariate normal distribution function, and Plackett's identity $\partial N_2/\partial\rho = \varphi_2$ with the symmetry $\varphi_2(-\kappa,-\kappa;\rho) = \varphi_2(\kappa,\kappa;\rho)$ give $\partial\varpi_\rho/\partial\rho = \varphi_2(\kappa,\kappa;\rho)$ on $(-1,1)$; at $\rho = 0$ this is $\varphi(\kappa)^2$. Since $\frac{d}{d\rho}h(\frac{1+\rho}{2}) = \frac{1}{2}\log\frac{1-\rho}{1+\rho}$ vanishes at 0,

$$\zeta'_{\alpha,\kappa}(0) = \alpha\,\frac{\varphi(\kappa)^2}{\Psi(\kappa)^2} \;>\; 0,$$

so $\zeta_{\alpha,\kappa}(\rho_0) \geq \rho_0\,\zeta'_{\alpha,\kappa}(0)/2 > 0$ for all small $\rho_0 > 0$; fix such a ρ_0 and set $\delta = \zeta_{\alpha,\kappa}(\rho_0)$. For the weighted counts, the diagonal term of $\mathbb{E}[\mathcal{Z}_f^2]$ is $2^N\mathbb{E}[f(X)^2]^m$, so either $\mathbb{E}[\mathcal{Z}_f^2] = \infty$ and the claim is trivial, or $f \in L^2(\varphi\,dx)$ and Mehler's expansion gives $\mathbb{E}[f(X)f(Y)] = \sum_{j\geq 0}\rho^j\langle f, \mathrm{He}_j\rangle^2/j!$, whose ρ-derivative at 0 is $(\int xf\varphi\,dx)^2$; for $f \geq 0$ supported on $[\kappa,\infty)$ with $\kappa \geq 0$ the integrand $xf(x)\varphi(x)$ is nonnegative, and it vanishes almost everywhere only if f vanishes almost everywhere off the origin, which $0 < \int f\varphi\,dx$ excludes; the argument then runs unchanged. □

The mechanism is the asymmetry: pairs of solutions at small positive overlap are exponentially favored over independent pairs at every margin, which is what the AMP-conditioned second moment of [9] is built to circumvent at $\kappa = 0$, and what any lower-bound proof must circumvent at every other margin. At $\kappa = 0$ the failure is classical (see the introduction of [9]); Lemma 6 extends it to every κ and, for $\kappa \geq 0$, to every weighted count. For $\kappa < 0$ the weighted closure fails: a window weight symmetric about the origin has zero linear Mehler coefficient, which is the next observation.

8.2 A capacity lower bound at negative margins

Proposition 7. *Let $\kappa < 0$ and let $\alpha_c^{\mathrm{sym}}(r) = \log 2/(-\log\mathbb{P}(|X| \leq r))$ denote the symmetric binary perceptron threshold at width r. For every $\alpha < \alpha_c^{\mathrm{sym}}(|\kappa|)$, with probability $1 - o(1)$ the margin-κ perceptron admits a solution, that is, $m_N(\kappa)/N \geq \alpha$.*

Proof. If $|\langle g_a, w\rangle|/\sqrt{N} \leq -\kappa$ for all a then in particular $\langle g_a, w\rangle/\sqrt{N} \geq \kappa$, so the symmetric solution set at width $|\kappa|$ embeds in the margin-κ solution set. The value $\alpha_c^{\mathrm{sym}}(r)$ is the first-moment threshold of Aubin, Perkins and Zdeborová [16]. Altschuler's Theorem 1.4 [19] states that for every width $K > 0$ and every $\varepsilon > 0$ there is $c > 0$ such that, when the rows $g_a/\sqrt{N}$ are independent $N(0, N^{-1}I_N)$ vectors and their number is below $\alpha_c^{\mathrm{sym}}(K)N - c\log N$, a binary w with $|\langle g_a, w\rangle|/\sqrt{N} \leq K$ for every a exists with probability at least $1 - \varepsilon$ for N large. Its proof uses the second-moment condition of Abbe, Li and Sly [18, Lemma 3.4] (the second-moment exponent is largest at overlap zero), which they prove for every width in their Section 4.7.1. At fixed $\alpha < \alpha_c^{\mathrm{sym}}(|\kappa|)$ the row count αN is eventually below $\alpha_c^{\mathrm{sym}}(|\kappa|)N - c\log N$, so a symmetric solution of width $|\kappa|$ exists with probability $1 - o(1)$. (The lognormal limit of Abbe, Li, and Sly [18] is proved for Rademacher disorder, and Perkins and Xu [17] treat the Gaussian case under a single-critical-point hypothesis, so neither alone covers the model here.) □

The bound should be read next to monotonicity. A solution at margin 0 is a solution at every $\kappa < 0$, so $m_N(\kappa) \geq m_N(0)$, and at $\kappa = 0$ the ratio $m_N(0)/N$ converges in probability to $\alpha_\star(0) \approx 0.8330786$ (the lower half by [9] with the sharp-threshold transfer of [21], its numerical hypotheses certified in [14]). Hence for $\kappa < 0$

$$\frac{m_N(\kappa)}{N} \;\geq\; \max\bigl\{\alpha_\star(0),\, \alpha_c^{\mathrm{sym}}(|\kappa|)\bigr\} - o(1) \quad \text{whp.}$$

At the two certified negative margins monotonicity is the stronger input: the embedding gives 0.215 at $\kappa = -0.05$ and 0.655 at $\kappa = -0.45$, against the predicted 0.889 and 1.558. The embedding takes over only for $\kappa < -0.5757$, where $\alpha_c^{\mathrm{sym}}(|\kappa|)$ passes $\alpha_\star(0)$. Li, Schramm, and Zhou [23] prove lower bounds on the satisfiability threshold of the same one-sided perceptron at negative margin by a different route, with the sharpest statements in the $\kappa \to -\infty$ regime.

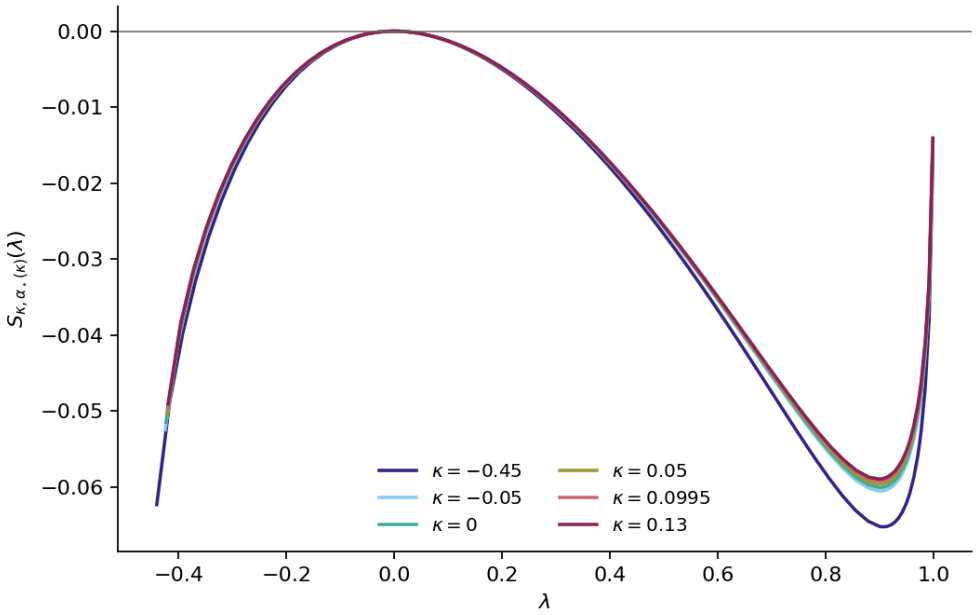


Figure 2: The general-margin Ding–Sun lower-bound condition, evaluated at the five certified margins and $\kappa = 0$: the univariate function $S_{\kappa,\alpha_\star(\kappa)}$ of their equation (48), whose nonpositivity off $\{0, 1\}$ is the numerical hypothesis of their second-moment argument. All six curves are negative off the two roots and nearly coincide.

8.3 The Ding–Sun condition at general margin

The construction underlying the $\kappa = 0$ lower bound of [9] is written at general margin throughout their Section 2: their equations (34)–(48) define, for parameters $(\kappa_{\mathrm{all}}, \alpha_{\mathrm{all}})$, an explicit univariate function $S_{\kappa,\alpha}(\omega)$ on $[\omega_{\min}, 1]$ built from the fixed point $(q_\star, \psi_\star)$ (an entropy term through the two-spin measure $P_{H,D}$, a constraint term through the conditional Mills law φ_ξ, and a convex $\inf_s$ correction), and their Condition 1.2 is the statement $S_{0,\alpha_\star}(\omega) < 0$ for $\omega \notin \{0, 1\}$. The κ-analog of that hypothesis is therefore already defined in their own notation, and its κ-dependent inputs are the fixed point, its uniqueness rectangle and the crossing, which are given by Theorem 5 and the enclosures of Theorem 4.

In floating point, with Gauss–Hermite and conditional Gauss–Legendre quadrature, the function $S_{\kappa,\alpha_\star(\kappa)}$ behaves uniformly at the five certified margins and at $\kappa = 0$ (Figure 2): at every margin the curve is strictly negative off $\{0, 1\}$, at most $-5 \cdot 10^{-5}$ outside windows of half-width 0.02 around the two roots (inside them it climbs to its maxima at the roots, reaching $-1.6 \cdot 10^{-6}$ at the closest grid points), the maximum at $\omega = 0$ is degenerate exactly as at $\kappa = 0$, and the whole family collapses nearly onto a single shape, with $\omega_{\min}$ drifting from -0.441 at $\kappa = -0.45$ to -0.421 at $\kappa = 0.13$. Certifying $S_{\kappa,\alpha_\star} < 0$ off the roots is a problem in one variable, simpler than the two-variable certificate of Section 6.

What separates these verified inputs from a general-margin lower bound is the probabilistic body of [9]: the AMP-conditioned moment computation of their Sections 2–7, whose displays carry κ_{all} explicitly; by their Remark 1.5 the assumption $\kappa = 0$ enters only in a few steps, which they indicate. Extending that body to $\kappa \neq 0$ requires, beyond the certified inputs above: a

priori bounds at margin (the upper one is free at every κ from the first moment, since $\mathbb{E}\mathcal{Z} \to 0$ exponentially above the annealed threshold $-\log 2/\log \Psi(\kappa)$; for $\kappa < 0$ the lower one is the monotonicity bound above), the state-evolution inputs (generic in the nonlinearity, hence in κ), and the sharp-threshold transfer of [21] if convergence in probability rather than positive probability is wanted.

9 The trusted base

Every inequality of Theorems 4 and 5 is decided by a comparison of arb balls (python-flint 0.9.0 on FLINT 3.6.0), built from the primitives of [14]: monotone endpoint evaluations, mean-value cells, correlated integrands and tail bounds. The identities the certificates use (Lemmas 1 and 2, the form of ℓ, the constraint kernels at general margin and the integrand written against b_0) are proved in the text.

The certificates of Condition 4 record every sign-critical quantity as an outward-rounded decimal in canonical JSON, bound by SHA-256 hashes; the stage-2 file binds the hash of the Region I file it reads. Their exact layer is restated over the integers and checked by the Lean 4 kernel (core library, v4.35.0): the star-interior inequality chain and the positivity of its perturbation determinant, the positive definiteness of the Loewner majorants at all five margins, and the gap-free tilings of the Region II and stage-2 grids and of the 1404-band Region I schedule. Two global clauses on the band geometry, that the innermost ring closes the full circle and that each level's angular support lies inside the previous level's, together make every radial chain contiguous out to the star radius. These two clauses are read off the certificate files directly; they and the analytic lemmas are outside the formal check.

The enclosures behind Theorem 5 are not restated in exact arithmetic; that theorem rests on the ball-arithmetic computation alone.

10 What the curve says about learning

For binary weights the Euclidean norm is fixed at $\sqrt{N}$, so a margin constraint is exactly a robustness certificate: if $y_a\langle w, x_a\rangle/\sqrt{N} \geq \kappa$ then the label y_a is invariant under every perturbation of x_a of Euclidean norm below κ. With centrally symmetric i.i.d. designs and independent labels, label folding makes the robust-interpolation event identical to the satisfiability event studied here, so $\alpha_\star(\kappa)$ is the predicted threshold for how many random labels per parameter a one-bit-per-parameter linear classifier can memorize with prescribed robustness radius. The certified $\kappa = 0.05$ bound, for instance, says such a classifier cannot robustly store more than 0.781075 random labels per parameter at that radius, with the guarantee holding against a worst-case adversary in feature space.

Read quantitatively, on the certified strip the logarithm of the threshold decreases at a rate that falls from about 1.55 per unit margin at $\kappa = -0.65$ to about 1.26 near $\kappa = 0.1$ (1.40 on average), so robust storage capacity halves for every 0.45 to 0.55 of demanded feature-space radius, and the decay continues smoothly beyond (a factor 4.8 from $\kappa = 0$ to 1.5). The efficiency of binary relative to real-valued weights *rises* with the demanded robustness: $\alpha_\star/\alpha_{\text{spherical}}$ grows from 0.417 at $\kappa = 0$ to 0.556 at $\kappa = 1.5$, so one-bit parameters give up proportionally less storage as the required robustness grows. On the relaxed side ($\kappa < 0$, tolerating margin violations up to $|\kappa|$) capacity grows toward the annealed bound (the

ratio reaches 0.996 by $\kappa = -2$) and the solution space approaches the replica-symmetry-breaking regime (the weak Hessian eigenvalue of the two-variable function degrades threefold by $\kappa = -1.5$), which is where the replica-symmetric prediction itself should eventually fail. The replica-symmetric cavity picture also predicts the margin distribution *among* solutions: conditionally on the quenched component $\sqrt{q}\,Z$ of a constraint's field, the field on a typical solution is $N(\sqrt{q}\,Z,\, 1-q)$ truncated to $[\kappa, \infty)$, so the gap density across constraints is the mixture $p(h) = \mathbb{E}_Z[\varphi_{1-q}(h - \sqrt{q}\,Z)\,\mathbf{1}\{h \geq \kappa\}/\Psi(u(Z))]$. Computed along the branch at $\kappa = 0.05$ (a float diagnostic, `margin_distribution.py`), the mean gap above the demanded margin compresses from 0.70 at $\alpha = \frac{1}{2}\alpha_\star$ to 0.59 at $\alpha_\star$: solutions are squeezed against the constraint boundary as storage saturates, which is the quantity a margin-type generalization argument would consume, and a natural next target for the certified machinery. These are statements about information-theoretic storage, not about algorithms: for the Ising perceptron the known efficient algorithms stall far below $\alpha_\star$, and the margin curve gives the ceiling against which that algorithmic gap should be measured at every κ. Limits of the reading: nothing here concerns generalization on structured data, correlated features, or hidden layers; κ is a feature-space radius, and translating it to input space requires a Lipschitz bound on the feature map.

The gap between the curve and what an algorithm finds can be measured at moderate size. A population annealer (hinge energy on violated margins, geometric schedule matched to the $2/\sqrt{N}$ energy scale; `finite_size_experiment.py`) trained $N = 101$ instances across α at $\kappa \in \{0, 0.05, 0.1\}$. The densities at which half of its runs succeed fall in the order of κ and sit about a tenth of $\alpha_\star(\kappa)$ below it at this budget, while scattered successes persist to the bound itself: at $\kappa = 0.1$ the last appear at $\alpha \approx 0.73$, essentially at $\alpha_\star$, and none beyond. The solutions the annealer does find are atypical in exactly the direction the clustering picture predicts: at low load their mean gap above κ tracks the typical-solution law $p(h)$, while near the half-success density it holds around 0.65–0.70 as the typical mean descends; the algorithm reaches the wide, over-margined minority. At $N = 201$ under the same per-spin budget the success curves sharpen and shift left, so what the annealer traces is a budget-dependent lower envelope of the true algorithmic threshold, whose location and the clustering barrier behind it are studied in [20, 23] on the closely related symmetric and asymmetric binary perceptrons; the found-solution histograms meanwhile track the derived $p(h)$ closely at moderate load and deform toward mid-gaps near the half-success density. Within each size and budget the half-success density sits at a κ-independent fraction of $\alpha_\star(\kappa)$ (0.90 at $N = 101$ down to 0.50 at $N = 801$, flat across margins to better than one percent at the largest sizes), so the demanded margin costs the algorithm the same relative capacity it costs the storage bound (Figure 3). At fixed sweeps per spin the fraction falls with N (0.90, 0.75, 0.69, 0.63, 0.50 at $N = 101$, 201, 301, 401, 801), close to a fixed loss of 0.13 per doubling over the measured range; holding a constant fraction of capacity requires superlinearly growing total work. On the budget axis at $N = 201$ the fraction climbs 0.75, 0.79, 0.86 as the budget quadruples twice, but a further quadrupling to 25,600 sweeps per spin gains only 0.02 (0.876), and the half-success density saturates near 0.88 of capacity at this size over the measured range.

The cost of a margin also depends on the weight alphabet. Solving the replica-symmetric system for ternary weights $\{-1, 0, +1\}$ with a self-overlap order parameter (`exploration/ternary_km.py`) gives $\alpha_\star^{\mathrm{tern}}(0) = 1.17448$ against an annealed bound of $\log_2 3$, with 1.07951 at $\kappa = 0.05$ and 0.99483 at $\kappa = 0.1$. Per weight *state* the richer alphabet stores more; per weight *bit* it stores less: 0.741 against the binary 0.833 constraints per bit at $\kappa = 0$, and the gap

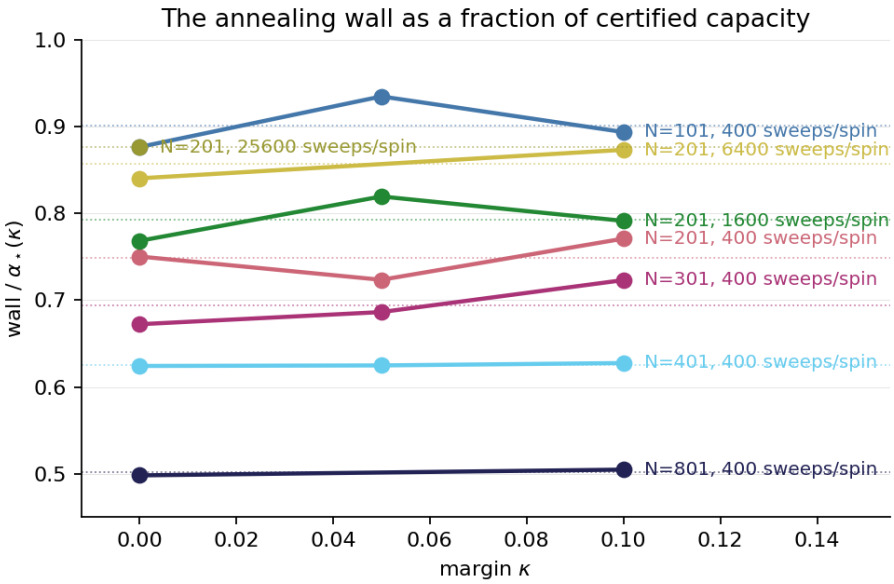


Figure 3: The half-success density of the annealer as a fraction of the predicted capacity $\alpha_\star(\kappa)$, across size–budget pairs. Within each run the fraction is flat in κ to a few percent, while its level depends on both size and budget.

widens monotonically with margin (0.100 at $\kappa = 0.05$, 0.105 at 0.1). The full eight-point curve (Figure 4, κ up to 0.35) makes this a statement about the whole margin range rather than three samples: per stored bit the binary curve sits above the ternary one everywhere, the absolute gap saturating near 0.11 constraints per bit above $\kappa \approx 0.2$ while the relative advantage grows from 12 to 26 percent, since both capacities fall with margin.

11 Scope and further directions

Other margins of the same family. The certificates of Section 6 can be run wherever the continuation reaches: positive margins to $\kappa \approx 1.18$ (past which the fixed-tilt center certificate must re-optimize the tilt, as measured in Section 5.1), and negative margins until replica symmetry breaks, where the natural target is the approach (the degradation of the weak eigenvalue and of the slack in ℓ_0) rather than the condition.

Other priors and activations. The moment-coordinate reduction needs only that the profile enter through finitely many moments of a log-concave base pair; replacing $\{-1, +1\}$ by a bounded discrete prior changes Φ and the moment body but not the architecture. The spherical perceptron at margin (Gardner's curve, shown for context in Figure 1) has simpler convexity structure and is a validation target where exact answers are classical.

Teacher–student and structured labels. The planted version replaces the annealed constraint law by a tilted one; the cavity argument and the two-variable function have direct analogues, and the certified-slab design (per-slab rectangles, located boxes, deep bisection) transfers whenever the fixed-point system does.

Toward learning theory. Two quantitative questions are natural: certified upper bounds

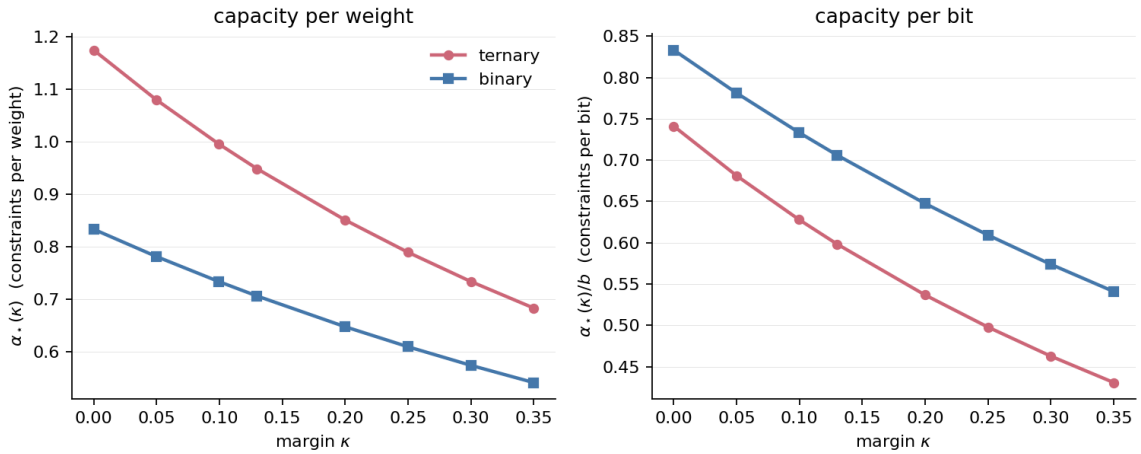


Figure 4: Replica-symmetric capacity of ternary against binary weights, per weight (left) and per stored bit (right, dividing by $\log_2 3$ and 1). The order reverses: the richer prior stores more per weight and less per bit, at every computed margin.

along capacity curves for one-bit compressed classifiers (the κ-robust memorization ceiling above, made uniform over the strip), and certified bounds on the margin distribution achievable at a given $\alpha < \alpha_\star(\kappa)$, which is the object that connects storage geometry to generalization bounds of the classical margin type.

The lower bound. Ding and Sun's conditioned second-moment argument is carried out at $\kappa = 0$; its hypothesis at general margin is the one-variable inequality $S_{\kappa,\alpha_\star} < 0$ of Section 8, which holds in floating point at the five margins.

Notation

The last column gives the section where the symbol is defined. Symbols used only inside one proof are not listed.

Symbol	Meaning	Section
N	number of binary weights	1
w	weight vector in $\{-1,+1\}^N$	1
g_a	a-th pattern, a standard Gaussian vector in $\mathbb{R}^N$	1
m	number of patterns, $m = \lceil \alpha N \rceil$	1
α	pattern density m/N	1
κ	margin	1
$\varphi,\ \Psi$	standard Gaussian density and upper tail	2
E	inverse Mills ratio φ/Ψ	2
Z	standard Gaussian variable	2
$u(z;q,\kappa)$	$(\kappa - \sqrt{q}\,z)/\sqrt{1-q}$	2
$q,\ \psi$	the two order parameters	2
$P,\ R$	the two halves of the fixed-point map, (1)	2

Symbol	Meaning	Section
G	replica-symmetric free energy, (2)	2
$G_\star(\alpha)$	G along the fixed-point branch	2
$\alpha_\star(\kappa)$	the zero of $G_\star$; the predicted capacity	2
$(q_\star, \psi_\star)$	the fixed-point branch as a function of α	2
(q_0, ψ_0)	the fixed point at $\alpha_\star(\kappa)$	2
$\ell(\hat{z}),\ \ell_0$	the local-concavity function of Condition 3 and its minimum	4.1
$\mu(\hat{z})$	$\mathbb{E}[(\hat{z} + \cosh^2(\sqrt{\psi_0}Z))^{-1}]$	4.1
$\tau,\ A,\ \beta$	$E'(u)$, $1 - q_0$ and $\mu(\hat{z}) - A$ in the formula for ℓ	4.1
$\mathcal{H}$	Hessian of Huang's first-moment functional at its maximizer	5.1
η	shifted cavity variable $E\big((\kappa - \hat{H})/\sqrt{1-q_0}\big)/\sqrt{1-q_0}$	5.1
$\dot{H},\ \hat{H}$	independent Gaussians of variances ψ_0 and q_0	5.2
M	$\tanh \dot{H}$	5.2
Λ	a profile, a function of $(\dot{H}, M)$ with values in $[-1, 1]$	5.2
$a = (a_1, a_2)$	the moments $(\mathbb{E}[\dot{H}\Lambda], \mathbb{E}[M\Lambda])$	5.2
K	the moment body, the set of attainable a	5.2
$\mathcal{S}(\Lambda, s)$	Huang's two-variable function at tilt s	5.2
$\mathcal{S}_\star(\lambda_1, \lambda_2)$	$\inf_{s \geq 0} \mathcal{S}(\tanh(\lambda_1 \dot{H} + \lambda_2 M), s)$	5.2
$\lambda,\ b$	dual points in $\mathbb{R}^2$	5.2
$\Phi(b)$	$\mathbb{E} \log 2\cosh(b_1 \dot{H} + b_2 M)$	5.2
$H(a)$	entropy term as a function of the moments	5.2
$a^\star$	the maximizer $(\psi_0(1 - q_0), q_0)$	5.2
s_0	the optimal tilt $\sqrt{1 - q_0}$ at $a^\star$	5.2
$\mathcal{F}(a, s)$	$\mathcal{S}(\cdot, s)$ written in moment coordinates	6
$v,\ c,\ t$	ray direction, tilt slope and distance along the ray	6
$\chi_{v,c}(t)$	$\mathcal{F}(a^\star + tv,\ s_0 + ct)$	6
Δ	$(\lambda_1 - 1)\dot{H} + \lambda_2 M$ in the integrand written against b_0	6
$V,\ \sigma$	cavity argument and $E(V) - V$	Lemma 3
B	interval majorant of $\nabla^2 \Phi$ on a Region I band	7
$m_N(\kappa)$	largest m for which a solution exists	8
$S_{\kappa,\alpha}(\omega)$	Ding and Sun's univariate function on $[\omega_{\min}, 1]$	8

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital.

Declaration of generative AI and AI-assisted technologies

The mathematics, the software, the certificates and the drafting of this paper were done by AI agents, Claude (Anthropic) and Codex (OpenAI), working through the shared memory of the Math Brain project: its code and tests [24], the framework it runs on [26] and its description [25]. The author set the objective, supervised the runs, reviewed the mathematics and accepts responsibility for the paper.

Data availability

No external data were used. The verification programs, certificates and raw records are available at `https://github.com/yspennstate/ising-perceptron-margin/tree/v2.0`.

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